

A phase-field model of elastic and viscoelastic surfaces in fluids

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ARTICLE INFO

Keywords:

Phase-field modeling
Diffuse interface modeling
Elastic shell
Viscoelastic surfaces
Two-phase flow

ABSTRACT

Elastic surfaces immersed in fluids arise in a wide range of scientific and engineering applications. As opposed to traditional numerical approaches based on a triangulation of the surface, we propose a phase-field model which incorporates surface elasticity, including in-plane stretching and shearing. The model is based on a phase-field model for two-phase flow coupled to a fully Eulerian description of the surface deformation tensor. We put a particular focus on purely dilational and 3D axisymmetric surfaces. For these cases we propose models to track the evolution of in-plane strains and stresses by simple scalar surface PDEs, which can be localized to the diffuse interface by use of an approximate surface delta function.

To validate the applicability and accuracy of the phase-field model we conduct several numerical experiments and show convergence towards reference solutions. Finally, we highlight some distinct advantages of the presented model, including the ability to handle topological changes, while ensuring conservative properties. Including relaxation terms, we extend the model to viscoelastic in-plane surfaces which enables to tackle a range of complicated problems such as surface fluidization and solidification in the future.

1. Introduction

Fluid–structure interaction (FSI) problems arise in a wide range of scientific and engineering applications, such as aeroelasticity, biological fluids and biomechanics. While traditional FSI problems focus on the interaction between *volumetric* solid bodies and fluids, recent years have seen an increasing interest in problems involving elastic *surfaces* in fluids. For example, the modeling and simulation of elastic surfaces is key to study the dynamics of capsules in fluid flow with numerous applications in food, cosmetics and pharmaceutical industries [1]. Also in biology, the interaction of fluids with biological membranes and cell surfaces has become a subject of intense study due to the growing recognition of the role of mechanics in the functioning and disfunctioning of biological systems [2–4]. In medicine, numerical simulations of elastic biological surfaces in fluid flow are nowadays even used to detect diseases [5–7].

Capsules, membranes and shells are in general modeled as liquid droplets protected by a thin elastic sheet. While the definition of elastic surface stresses has been formalized decades ago [8,9], different approaches exist to track the position of the elastic surface. Traditionally a triangulated grid is used that follows the displacement of the interface. One of the most classical approaches of this kind is the Arbitrary Lagrangian–Eulerian (ALE) method [10–13]. This approach is based on a surface-fitted finite-element grid, where surface grid points are shared with the surrounding fluid mesh, which in turn moves consistently with surface deformations. While this approach leads to very accurate results, its limitations are clear when dealing with large deformations or contact problems, which lead to distortion or rupture of the fluid mesh. Another approach is the immersed boundary method where different grids

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are used to represent either the fluid or the embedded surface [14] and their coupling is realized by smeared-out Dirac delta functions. However, the insertion or deletion of grid points during large stretching and shrinking of the membrane remains a delicate aspect [15]. Moreover, as both ALE and immersed boundary methods are based on an explicit triangulation of the surface, scenarios which involve contact (e.g. surface-wall contact) or topological changes are excluded.

These problems have led to the development of alternative modeling approaches in recent years, in particular fully Eulerian formulations and interface capturing methods. In interface capturing methods, the surface is implicitly described by an additional field variable. Most popular interface capturing methods are the level-set [16], volume-of-fluid [17], and phase-field method [18]. A pioneering approach to include full surface elasticity in interface capturing methods was presented in [19], based on a level-set description of the interface. Despite being numerically improved in [15] the framework still suffers from the need of artificial correction terms (e.g. displacement diffusion, displacement extension, mass conservation) which introduce additional numerical errors into the simulations, accumulating over time.

In this work we propose an alternative formulation based on a diffuse interface description of the surface by a phase-field. Depending on the application of interest, phase-field methods offer some distinct advantages over other interface capturing methods. For example, they can intrinsically include mass conservation and transport stabilization and allow for fully discrete energy stable schemes, see e.g. [20,21], for two-phase flows. Moreover, a linear implicit coupling of phase-field advection and flow equations can prevent time step restrictions due to stiff interfacial forces [22]. Volumetric elasticity has been included in phase-field models by advecting a strain tensor [23] or a reference map [24]. Surface elasticity, on the other hand, has only been included in phase field models to account for the out-of-plane component, namely bending stiffness [25,26]. The in-plane elastic contributions, namely stretching, compression, and shearing, have so far been neglected. One notable exception is the inclusion of surface inextensibility in [27], which can be used to approximate the behavior of lipid bilayer membranes in a limited parameter range.

In this paper we aim to add the in-plane component of elastic surfaces to phase-field models. Similar to the phase-field models of volumetric FSI [23], we base the derivation on an evolution equation for elastic stresses or strains, which we introduce in the sharp-interface setting (Section 2.1). After discussing the extension to viscoelastic surfaces (Section 2.2), we present the well-known diffuse interface model of bending elasticity. When adding the in-plane elasticity, we put a particular focus on two cases: purely dilational surface elasticity (Section 3.2) and 3D axisymmetric surface elasticity (Section 3.3). Axisymmetry not only reduces the computational costs but also enables a strongly simplified definition of the surface stress. Moreover, we show that the developed PDEs can be formulated very elegantly in a diffuse interface fashion, while maintaining conservation properties. Afterwards, we provide the basic idea for a full 3D model (Section 3.4). The model is discretized by a finite-element method (Section 4) and validated in a series of numerical tests (Sections 5.1 and 5.2). Finally, we highlight some distinct advantages of the new model as compared to traditional interface tracking methods. To this end, we simulate an elastic surface traversing a microfluidic channel without re-meshing (Section 5.3) and the topological change at the fusion of two vesicles (Section 5.4). The paper closes with a conclusion on the applicability of the method in Section 6.

2. Sharp interface model

Before deriving a diffuse interface model of elastic surfaces in fluid flow, we denote the corresponding sharp-interface equations.

2.1. Sharp-interface model of an elastic surface in fluid flow

Throughout this work we see the following sharp interface setting as the ground truth to describe a two-phase flow problem of an elastic membrane and aim to approximate it using a diffuse interface approach. Within the sharp interface model we consider a time dependent elastic surface Γ within a domain $\Omega \subset \mathbb{R}^d$, $d = 2, 3$. The closed surface Γ separates the domain into the external fluid Ω_0 and the internal fluid Ω_1 , with $\Omega = \Omega_0 \cup \Omega_1 \cup \Gamma$ (see Fig. 1). We model the flow of the immiscible fluids using the incompressible Navier–Stokes equations. Let $\mathbf{v} : \Omega \rightarrow \mathbb{R}^d$ be a continuous velocity field to indicate the movement of the fluids. The balance of laws for mass and momentum give

$$\rho_i \partial_t^* \mathbf{v} - \nabla \cdot (\eta_i (\nabla \mathbf{v} + (\nabla \mathbf{v})^T)) + \nabla p_i = 0 \quad \text{in } \Omega_i, i = 0, 1, \quad (1)$$

$$\nabla \cdot \mathbf{v} = 0 \quad \text{in } \Omega_i, i = 0, 1, \quad (2)$$

where ∂_t^* denotes the material derivative and ρ_i, η_i, p_i are the density, viscosity and pressure in the two fluids, respectively. At the interface fluid forces are balanced by interfacial forces, i. e.,

$$[\eta_i (\nabla \mathbf{v} + (\nabla \mathbf{v})^T) - p_i \mathbf{I}]_0^1 \cdot \mathbf{n} = \frac{\delta E_{tension}}{\delta \Gamma} + \frac{\delta E_{bend}}{\delta \Gamma} + \frac{\delta E_{stretch}}{\delta \Gamma} \quad \text{on } \Gamma. \quad (3)$$

Here $[\cdot]_0^1$ denotes the jump of a quantity across the interface and $\mathbf{n}$ the normal vector to Γ . The right hand side represents all interfacial forces to be considered within this work, formulated as variational derivative of corresponding energies.

The surface tension energy $E_{tension}$, which tends to minimize the surface area, reads

$$E_{tension} = \int_{\Gamma} \sigma_0 dA, \quad (4)$$

with a material specific surface tension σ_0 , assumed to be constant on the whole surface Γ . The membranes tendency to assume a preferred curvature is described by the bending energy E_{bend} , which tends to minimize the deviation of the total curvature κ from

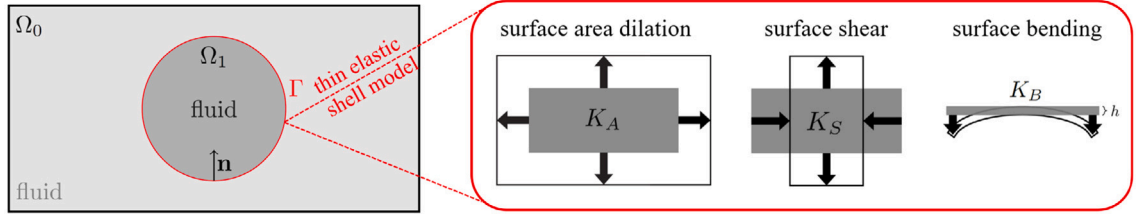


Fig. 1. Left: Illustration of the domain $\Omega = \Omega_0 \cup \Omega_1 \cup \Gamma$ for a two-phase flow problem of an elastic shell Γ in a sharp interface setting. Right: Illustration of the mechanical thin shell model with the three distinct elastic moduli. (K_A) The surface element (gray) is stretched with constant aspect ratio and increasing area. (K_S) The surface element is stretched with constant area but nonconstant aspect ratio. (K_B) Out-of-plane bending deformation of the surface element with thickness h (lateral view).

the material's reference curvature κ_0 . Assuming a Hookean response of the membrane to bending as done by Canham [28] and Helfrich [29] the bending energy is given by

$$E_{bend} = \int_{\Gamma} \frac{1}{2} K_B (\kappa - \kappa_0)^2 dA, \quad (5)$$

where K_B is the bending stiffness.

Finally, assuming the surface to be an isotropic thin shell of thickness $h \ll 1$ leads to the stretching energy $E_{stretch}$ which tends to minimize in-plane stretching and compression of the surface compared to its reference state. Various forms of $E_{stretch}$ are available. Assuming a neo-Hookean material response [30], the energy is

$$E_{stretch} = \int_{\Gamma} \frac{K_A - K_S}{2} (J - 1)^2 + \frac{K_S}{2} (\lambda_1^2 + \lambda_2^2 - 2 - 2 \ln J) dA. \quad (6)$$

Here K_A and K_S are the surface areal (dilatational) modulus and shear modulus, respectively. The principle stretches λ_1 and λ_2 give the relative change in length (i. e. $\lambda = l/l_0$) along the two tangential, orthogonal directions along which the deformation is maximal in-plane to the surface. The local area stretch $J := \lambda_1 \lambda_2$ gives the ratio of the current area of a surface element divided by the initial area of the same element, $J = A/A_0$, which means $J = 1$ in the reference state.

Let us note that the surface parameters K_A , K_S and K_B can be computed from the 3D mechanical parameters of the surface material. In 3D elasticity theory, the response of an isotropic elastic body to elastic deformations is described by the Young's modulus E and the Poisson's ratio ν . For an elastic thin shell of thickness h , it holds

$$K_A = \frac{Eh}{2(1-\nu)}, \quad K_S = \frac{Eh}{2(1+\nu)}, \quad K_B = \frac{Eh^3}{24(1-\nu^2)}, \quad (7)$$

see e. g. [31]. The effect of these contributions is also displayed in Fig. 1. The area dilation modulus K_A describes the response of the surface to area changes (in-plane) with constant aspect ratio. The response to in-plane shear deformations with constant area of the surface element is described by the area shear modulus K_S . Finally, the bending stiffness (also referred to as the bending modulus) K_B describes the resistance of the surface to out-of-plane bending deformations.

2.2. Sharp interface model of a viscoelastic surface

In particular in biological applications, many surfaces possess both elastic and viscous properties. The theory of viscoelasticity is divided into two fundamentally different cases: viscoelastic fluids (Maxwell type) and viscoelastic solids (Kelvin-Voigt type). While viscoelastic *solid* surfaces can be realized by simply adding a viscous term to the previously given surface forces, viscoelastic *fluid* surfaces are more complicated to handle, since the elastic reference state changes over time as elastic stresses are dissipated. Consequently, an evolution equation for the viscoelastic stress is necessary.

A general constitutive model of a viscoelastic surface stress S was developed in [32] in a sharp interface setting. Correspondingly, the viscoelastic in-plane force is given by the surface divergence of this stress, i. e., $\mathbf{f}_{stretch} = \nabla_{\Gamma} \cdot S$. Following [32, see esp. Eqs. (29) and (30)] the stress is defined by the evolution

$$\partial_t^* S = \nabla_{\Gamma} \mathbf{v} S + S \nabla_{\Gamma} \mathbf{v}^T + 2K_S D_{\Gamma} + (K_A - K_S) \text{tr}(D_{\Gamma}) P - \frac{1}{\tau_S} S + \frac{\tau_A - \tau_S}{2\tau_A \tau_S} \text{tr}(S) P \quad \text{on } \Gamma, \quad (8)$$

where τ_S and τ_A are the surface shear relaxation time and the areal relaxation time respectively. Moreover, $P = I - \mathbf{n} \otimes \mathbf{n}$ denotes the surface projection matrix and $D_{\Gamma} = \frac{1}{2} (\nabla_{\Gamma} \mathbf{v} + (\nabla_{\Gamma} \mathbf{v})^T)$ the surface rate of deformation tensor. Based on this evolution, we will develop diffuse interface formulations of elastic and viscoelastic fluid surfaces in the following sections.

3. Diffuse interface model

3.1. Phase-field model for elastic surfaces in flow

In the described sharp interface model, the interface between the two fluids is considered an evolving free boundary Γ that can be tracked explicitly at any time. When using the phase-field method, an auxiliary function $\phi : \Omega \rightarrow [0, 1]$, the phase-field, is

introduced and the interface position is implicitly described by a level-set of ϕ instead. In our case this function takes the values $\phi = 0$ and $\phi = 1$ in the different fluid phases with a smooth transition in between, around the interface. Hence, the interface is diffuse with a finite width ϵ and can be located by an intermediate level-set of the phase-field (here $\phi = 0.5$).

The tension and bending energies can be reformulated into the diffuse interface approximations (see e.g. [33,34]),

$$\tilde{E}_{tension} = \sigma_0 \int_{\Omega} \left(\frac{\epsilon}{2} |\nabla \phi|^2 + \frac{1}{\epsilon} W(\phi) \right) dA, \quad (9)$$

$$\tilde{E}_{bend} = K_B \int_{\Omega} \frac{1}{2\epsilon} \left(\frac{1}{\epsilon} W'(\phi) - \epsilon \Delta \phi \right)^2 dA. \quad (10)$$

Here, $W(\phi)$ denotes a double-well potential which we choose as $W(\phi) = 18\phi^2(1-\phi)^2$ to eliminate any scaling factors in these energies. A similarly elegant phase-field approximation of the stretching energy $E_{stretch}$ cannot be constructed, since information on tangential stretching is not intrinsically included in the phase-field itself. Therefore, the inclusion of tangential stretching is the main goal of this paper.

The phase-field formulation of the hydrodynamic equations in Ω reads

$$\rho(\phi) \partial_t^* \mathbf{v} - \nabla \cdot (\eta(\phi) (\nabla \mathbf{v} + (\nabla \mathbf{v})^T)) + \nabla p = -\epsilon \nabla \cdot (\sigma_0 \nabla \phi \otimes \nabla \phi) + K_B g \nabla \phi + \mathbf{f}_{stretch}, \quad (11)$$

$$\nabla \cdot \mathbf{v} = 0, \quad (12)$$

where

$$g := \frac{1}{K_B} \frac{\delta \tilde{E}_{bend}}{\delta \phi} = \frac{1}{\epsilon^2} \mu W''(\phi) - \Delta \mu, \quad (13)$$

$$\mu := \frac{1}{\sigma_0} \frac{\delta \tilde{E}_{tension}}{\delta \phi} = \frac{1}{\epsilon} W'(\phi) - \epsilon \Delta \phi. \quad (14)$$

The viscosity $\eta(\phi)$ and the density $\rho(\phi)$ are interpolations of the corresponding values in the two fluids, chosen here as

$$\eta(\phi) = \eta_0(1-\phi) + \eta_1\phi, \quad \rho(\phi) = \rho_0(1-\phi) + \rho_1\phi.$$

While the forces from tension and bending are clear and well studied in the context of phase-field models, the in-plane elastic force $\mathbf{f}_{stretch}$ has not been formulated. The derivation of $\mathbf{f}_{stretch}$ for various scenarios is considered in the next sections of this paper.

Finally, the evolution of the phase-field must be prescribed to ensure that the interface moves consistently with the fluid flow. Different evolution equations are possible to achieve this. In the absence of in-plane stretching, a thermodynamic consistent system is given by

$$\partial_t^* \phi = M \Delta (\sigma_0 \mu + K_B g) \quad \text{in } \Omega, \quad (15)$$

where M is a mobility constant that scales with $\mathcal{O}(\epsilon)$ to approximate the correct sharp interface limit [35]. However, if neither bending stiffness nor surface tension are present (i. e. $K_B = \sigma_0 = 0$), the above phase-field evolution degenerates to a pure convection problem, which does not conserve the thickness of the interface. In our numerical tests we aim to test the later constructed stretching forces in isolation. In these cases we will rather use an advected Cahn–Hilliard equation, which does not degenerate for $K_B = \sigma_0 = 0$, hence,

$$\partial_t^* \phi = M \Delta \mu \quad \text{in } \Omega. \quad (16)$$

Note that this approach is consistent in the sense that the modeling error with respect to the sharp-interface model is unaffected. In particular the same sharp interface limit is reached.

3.2. Including dilational elasticity of the surface

We start with the simple case of a surface which exhibits an elastic response only for local changes in surface area. That is, shear elasticity of the surface is neglected. Prototypical examples of such surfaces are biological membranes whose constituents (lipid molecules) may freely rearrange in the tangential direction. Hence, tangential shearing of the surface is possible, but does not lead to an elastic restoring force. Also, surfaces in any two-dimensional simulation fall into this category as these cannot experience shear deformation due to the one-dimensional nature of the surface.

We start by deriving a model of dilational elasticity by introducing an evolution equation for the surface stretching in the sharp interface framework. The resulting surface equation will then be transformed into the diffuse interface context to yield a phase-field model of a stretchable elastic surface immersed in fluids. Finally, we give an alternative derivation, that extends the model to viscoelastic surfaces as well.

3.2.1. Derivation of sharp interface strain evolution

To include dilational elasticity into the model, requires to derive a form of the stretching force $\mathbf{f}_{stretch}$ which is suitable to be translated into the phase-field framework. Assuming a neo-Hookean response to surface compression and stretching, but no elastic response to shearing (i. e. $K_S = 0$), the stretching energy (6) simplifies to

$$E_{stretch} = \int_{\Gamma} \frac{1}{2} K_A (J - 1)^2 dA, \quad (17)$$

where K_A is again the area dilation modulus and J measures the surface dilation. The above energy was already present in the famous paper of Helfrich [29] and leads to an elastic response which is equivalent to the surface tension force

$$\mathbf{f}_{stretch} = \nabla_\Gamma \cdot (\sigma P) \quad (18)$$

$$\sigma = K_A(J - 1), \quad (19)$$

see [34]. The transformation of this force into the phase-field framework is straightforward if J is known. In mesh-based models, J can be computed by comparing the current positions of grid points of the surface mesh with their initial positions. In a phase-field model, however, such a direct computation of J will not be possible.

Instead, we propose to determine J from an evolution equation, which can be transferred to the phase-field context later. The change of the surface stretch is determined by the surface evolution equation

$$\partial_t^* J = J \nabla_\Gamma \cdot \mathbf{v} \quad \text{on } \Gamma, \quad (20)$$

where $\nabla_\Gamma \cdot := P : \nabla$ denotes the surface divergence operator. Note, that applying surface differential operators to a field variable requires the latter to be defined not only on Γ but in a neighborhood of Γ . Throughout this work, we assume all surface variables to be defined this way.

It is a-priori unclear how to transform Eq. (20) into the phase-field framework in an elegant and accurate way. To achieve this, let us rewrite the equation in terms of an auxiliary field variable c , which we define as the inverse of the stretch, $c = 1/J$. Hence, we can rewrite Eq. (20) and using the rules of differentiation ($\partial_t^* c = -\frac{1}{J^2} \partial_t^* J$) we obtain the conservative evolution

$$\partial_t^* c + c \nabla_\Gamma \cdot \mathbf{v} = 0 \quad \text{on } \Gamma. \quad (21)$$

Note that this is the prototype of the evolution equation for a conserved surface-bound concentration field. Hence, c may be interpreted as the concentration of surface material (e. g. lipid molecules) per unit area. This kind of surface equation can be elegantly treated with the diffuse interface approach as we will do in the next section. Given that the surface is initially in the elastic reference state, i. e. $J = 1$, leads to the initial condition $c = 1$ on Γ . From Eq. (19) we obtain $\sigma = K_A \left(\frac{1-c}{c} \right)$. This elastic surface tension can be added to σ_0 in the momentum balance equation (11) to provide the elastic in-plane force in the diffuse interface framework.

3.2.2. Diffuse interface approximation

The advantage of the above formulation with a concentration field c is that conservative evolution equations can be restricted to a diffuse interface in a very elegant way. To illustrate the idea, consider the weak formulation of Eq. (21). Testing with a suitable test function ψ with compact support in $\Omega \times [0, T]$ we obtain

$$\begin{aligned} 0 &= \int_\Gamma \psi [\partial_t^* c + c \nabla_\Gamma \cdot \mathbf{v}] \, dx = \int_\Gamma \partial_t^* (c\psi) - c \partial_t^* \psi + c \psi \nabla_\Gamma \cdot \mathbf{v} \, dx \\ &= \frac{d}{dt} \int_{\Gamma(t)} \psi c \, dx + \int_\Gamma -c \partial_t^* \psi \, dx, \end{aligned}$$

where Reynolds transport theorem on surfaces was used in the last step. Now the domain of integration can be extended to $\Omega \supset \Gamma$ by multiplying the integrand with an interface Dirac delta function. Approximating the delta function by $|\nabla\phi|$ leads to

$$0 = \frac{d}{dt} \int_\Omega |\nabla\phi| \psi c \, dx + \int_\Omega -|\nabla\phi| c \partial_t^* \psi \, dx.$$

Using the ordinary Reynolds transport theorem (not on a surface) yields

$$0 = \int_\Omega \partial_t^* (|\nabla\phi| c) \psi + |\nabla\phi| c \psi \nabla \cdot \mathbf{v} \, dx = \int_\Omega \partial_t (|\nabla\phi| c) \psi + \nabla \cdot (|\nabla\phi| c \mathbf{v}) \psi \, dx.$$

Hence, we have reformulated the surface equation to a simple PDE in Ω

$$\partial_t (|\nabla\phi| c) + \nabla \cdot (|\nabla\phi| c \mathbf{v}) = 0. \quad (22)$$

However, the diffuse interface formulation does not ensure that c is constant across the interface. In fact the sharp interface limit of Eq. (22) does not even guarantee that c is continuous across Γ , which can lead to several problems when including surface elasticity based on c . Therefore it is desirable to adapt the formulation such that c is continuous and even constant in normal direction across the interface, $\mathbf{n} \cdot \nabla c = 0$. To do so, we add a diffusion in normal direction, leading to the evolution

$$\partial_t (|\nabla\phi| c) + \nabla \cdot (|\nabla\phi| c \mathbf{v}) = D \nabla \cdot (|\nabla\phi| \nabla c \cdot \mathbf{n}_\phi \otimes \mathbf{n}_\phi) \quad \text{in } \Omega, \quad (23)$$

where $\mathbf{n}_\phi = \nabla\phi/|\nabla\phi|$ is a diffuse interface representation of the surface normal. Note, that the additional diffusion term with diffusion constant D does not interfere with the solution as it only acts in the direction normal to the interface. Asymptotic analysis shows that the diffusion term maintains the correct sharp interface limit, while ensuring that the concentration field remains continuous and constant in normal direction across the interface (see Appendix A).

Remark (Conservation of Elastic Reference State). The formulation of dilational surface elasticity based on Eq. (23) satisfies the conservation property $\int_\Omega |\nabla\phi| c = \text{const.}$, which can be seen by testing the weak form with $\psi = 1$. Since $c = 1/J$ and $\int_\Omega |\nabla\phi| c \approx \int_\Gamma c$,

this property approximates the identity $\int_{\Gamma} 1/J = \text{const.}$, which means that the surface area of the elastic reference state (the left-hand side) is constant. Hence, this desirable property is maintained in the phase-field formulation. We will introduce a conservative discretization scheme for Eq. (23) in Section 4 that maintains this property, thereby ensuring that the elastic reference state is maintained, even on the fully discrete level.

To sum up, we arrive at the following diffuse interface description of a two-phase flow with a dilationally elastic surface

$$\partial_t^* \phi = M \Delta \mu \quad (24)$$

$$\mu = \frac{1}{\epsilon} W'(\phi) - \epsilon \Delta \phi \quad (25)$$

$$\nabla \cdot \mathbf{v} = 0 \quad (26)$$

$$\rho(\phi) \partial_t^* \mathbf{v} - \nabla \cdot (\eta(\phi) (\nabla \mathbf{v} + (\nabla \mathbf{v})^T)) + \nabla p = -\epsilon \nabla \cdot (\sigma(c) \nabla \phi \otimes \nabla \phi) + K_B g \nabla \phi \quad (27)$$

$$\sigma(c) = K_A \frac{1-c}{c} + \sigma_0 \quad (28)$$

$$\partial_t (|\nabla \phi| c) + \nabla \cdot (|\nabla \phi| \mathbf{v} c) = D \nabla \cdot (|\nabla \phi| \nabla c \cdot \mathbf{n}_\phi \otimes \mathbf{n}_\phi) \quad (29)$$

$$g = \frac{1}{\epsilon^2} \mu W''(\phi) - \Delta \mu \quad (30)$$

in the whole domain Ω . Note, that a possible material surface tension σ_0 and bending stiffness force have been included for completeness (gray).

In the absence of bending stiffness, the system is very similar to a two-phase fluid with surfactants [36]. The main difference being the normal-diffusion in Eq. (29) and the different definition of the surface tension. In particular Eq. (29) gives a simple evolution equation to obtain the surface dilational stress of the system.

3.2.3. Extension to viscoelastic surfaces

The presented system can be easily extended to viscoelastic surfaces. In the absence of surface shear stresses ($K_S = 0$), the general model of viscoelastic fluid surfaces in Eq. (8) can be significantly simplified. In particular, only the trace of the viscoelastic stress tensor becomes relevant, as its traceless (deviatoric) part must vanish. As a consequence the viscoelastic surface force reduces to $\nabla_{\Gamma} \cdot \mathbf{S} = \nabla_{\Gamma} \cdot (\frac{1}{2} \text{tr}(\mathbf{S}) \mathbf{P})$, i. e., it is equivalent to a surface tension force with tension $\sigma = \frac{1}{2} \text{tr}(\mathbf{S})$.

Taking the trace of Eq. (8) we arrive at

$$\partial_{t,\Gamma}^* \text{tr}(\mathbf{S}) = 2K_A \nabla_{\Gamma} \cdot \mathbf{v} + \text{tr}(\mathbf{S}) \nabla_{\Gamma} \cdot \mathbf{v} - \frac{1}{\tau_A} \text{tr}(\mathbf{S}) \quad \text{on } \Gamma, \quad (31)$$

where τ_A is the only remaining viscoelastic relaxation time.

Again, we use an auxiliary field c to make this equation more amenable to a later transformation to the phase-field description. This time we define

$$c := \frac{1}{1 + \frac{\text{tr}(\mathbf{S})}{2K_A}}.$$

Using this to replace $\text{tr}(\mathbf{S})$ by c in Eq. (31) leads after a short calculation to the surface evolution equation

$$\partial_t^* c + c \nabla_{\Gamma} \cdot \mathbf{v} = \frac{1}{\tau_A} c(1 - c) \quad \text{on } \Gamma. \quad (32)$$

Analogously to Section 3.3.2 we can derive a diffuse interface approximation of Eq. (32), and obtain

$$\partial_t (|\nabla \phi| c) + \nabla \cdot (|\nabla \phi| \mathbf{v} c) = D \nabla \cdot (|\nabla \phi| \nabla c \cdot \mathbf{n}_\phi \otimes \mathbf{n}_\phi) + \frac{1}{\tau_A} |\nabla \phi| c(1 - c). \quad (33)$$

which replaces Eq. (29) for finite relaxation times. As a result, the system of Eqs. (24)–(28), (33), (30) describes a two-phase flow with a dilationally viscoelastic fluid surface. For $\tau_A \rightarrow \infty$ the model equation converges to the previous system of purely elastic surfaces, as expected. Therefore, the elastic surface model is derived from two fundamentally different starting points (elastic energy vs. viscoelastic stress evolution), which confirms its validity.

3.3. 3D axisymmetric elastic surfaces

After deriving the model for a purely dilational surface, we now consider general elastic surfaces, i. e., we include the shear elasticity. We start off with the axisymmetric case. This case is relevant and even omnipresent in applications at the small scale, at which the dominance of capillary forces leads to a spherical ground state of most shell-like objects [13].

3.3.1. Derivation of the sharp interface model

The restriction to axisymmetry does not only lead to an enormous reduction in computational costs, but also enables a completely different and strongly simplified definition of the elastic surface stress. The general mathematical definition of surface shearing involves a tangential surface tensor (S in Eq. (8)), to track the two principle stretches λ_1, λ_2 together with their two principle directions, see Section 2.1. Instead, for an axisymmetric surface, the principle directions are known to be radial and lateral. Hence, axisymmetric surface deformations can be described by tracking the two scalar quantities λ_1 and λ_2 only. These quantities are defined to measure the relative length change of a surface element in lateral (λ_1) and rotational (λ_2) direction, as illustrated in Fig. 2:

$$\lambda_1 = \frac{ds}{ds_0}, \quad \lambda_2 = \frac{r}{r_0},$$

where s is the arc length, r the distance to the symmetry axis, and s_0 and r_0 are the corresponding quantities at the same material point in the undeformed state [7].

To be consistent with previously used Hookean formulations [7,37,38], the surface stretch energy (6) can be approximated by a second-order polynomial expansion around the reference state ($\lambda_1, \lambda_2 \approx 1$) to obtain

$$E_{stretch} = \int_{\Gamma} \frac{1}{2} (K_A + K_S) ((\lambda_1 - 1)^2 + (\lambda_2 - 1)^2) + (K_A - K_S)(\lambda_1 - 1)(\lambda_2 - 1) dA. \quad (34)$$

The variational derivative of this energy w. r. t. surface movement gives the elastic force as previously computed in [7]:

$$\mathbf{f}_{stretch} = -\frac{\delta E_{stretch}}{\delta \Gamma} = \nabla_{\Gamma} \cdot (\sigma P) + 2K_S \begin{pmatrix} 0 \\ 1/r \end{pmatrix} (\lambda_1 - \lambda_2). \quad (35)$$

$$\sigma = (K_A + K_S)(\lambda_1 - 1) + (K_A - K_S)(\lambda_2 - 1). \quad (36)$$

Note, that again an effective elastic surface tension σ is part of the force.

It remains open how to track the principle stretches on an implicitly given surface in the phase-field framework. To do so, we acknowledge that axisymmetric problems are computationally solved in a two-dimensional cross section of the cylindrical domain. Let us call this cross-section Ω_{2D} and the contained surface contour $\Gamma_{1D} := \Gamma \cap \Omega_{2D}$. Without loss of generality, we assume the x -axis to be the axis of rotation such that the rotation of Γ_{1D} around the x -axis gives the full two-dimensional, axisymmetric hyper-surface Γ (see Fig. 2).

For the sake of brevity we abuse notation and define all fields only in Ω_{2D} in the following. The velocity field in this domain is defined to consist of only an axial and radial component $\mathbf{v} = (v_x, v_r)$. The projection matrix P becomes a 2×2 matrix containing only x and r components. Equally, the surface gradient and surface divergence operator in this domain are trivially defined by

$$\nabla_{\Gamma} \mathbf{v} := P \begin{pmatrix} \partial_x v_x & \partial_r v_x \\ \partial_x v_r & \partial_r v_r \end{pmatrix}, \quad \nabla_{\Gamma} \cdot \mathbf{v} := P : \begin{pmatrix} \partial_x v_x & \partial_r v_x \\ \partial_x v_r & \partial_r v_r \end{pmatrix}. \quad (37)$$

From the previous section we already found that the solution of a surface convection equation for a concentration c can be used to compute the surface dilation $J = 1/c$. Combining this with the fact that $J = \lambda_1 \lambda_2$ (see definition) leads to

$$\lambda_1 \lambda_2 = 1/c. \quad (38)$$

We can compute c by rewriting its evolution equation (21) into axisymmetric form, adding the axisymmetric contribution to the divergence operator

$$\partial_t^* c + c \left(\nabla_{\Gamma} \cdot \mathbf{v} + \frac{1}{r} v_r \right) = 0, \quad \text{on } \Gamma_{1D}. \quad (39)$$

On the other hand, if we omit the axisymmetric contribution (v_r/r), we effectively solve the evolution on the 1-dimensional interface Γ_{1D} instead of Γ . Following the argumentation in Section 3.2.1, the solution of this must provide the inverse of the lateral stretch of Γ_{1D} . Accordingly we define $c_{1D} = 1/\lambda_1$ and compute it by solving

$$\partial_t^* c_{1D} + c_{1D} \nabla_{\Gamma} \cdot \mathbf{v} = 0 \quad \text{on } \Gamma_{1D}, \quad (40)$$

Using Eq. (38) we conclude that by solving a set of two concentration equations we are able to obtain the two principal stretches:

$$\lambda_1 = \frac{1}{c_{1D}}, \quad \lambda_2 = \frac{c_{1D}}{c}. \quad (41)$$

After determining the stretches from c, c_{1D} , the elastic force can be computed using Eq. (35).

Remark. The presented ideas can be straightforwardly adapted to account for different material laws where the stress is constructed from λ_1, λ_2 (e. g. hyperelastic neo-Hookean materials).

3.3.2. Diffuse interface approximation

Similar to Section 3.2.2, the conservative evolution equations for the two concentrations can be restricted to a diffuse interface in a very elegant way, maintaining conservation properties. To obtain a diffuse interface formulation of the c_{1D} -Equation, we can use the same methodology as in Section 3.2.2 (only change Γ to Γ_{1D} and Ω to Ω_{2D}). As a result, we obtain

$$\partial_t (|\nabla \phi| c_{1D}) + \nabla \cdot (|\nabla \phi| \mathbf{v} c_{1D}) = D \nabla \cdot (|\nabla \phi| \nabla c_{1D} \cdot \mathbf{n}_{\phi} \otimes \mathbf{n}_{\phi}) \quad \text{in } \Omega_{2D}. \quad (42)$$

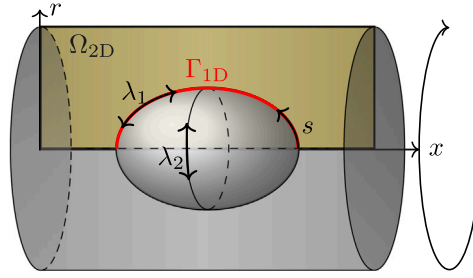


Fig. 2. Illustration of the two principal stretches of an axisymmetric elastic surface.

The axisymmetric equation for c (Eq. (39)) can be reformulated into

$$\partial_t^*(rc) + rc \nabla_\Gamma \cdot \mathbf{v} = 0 \quad \text{on } \Gamma_{1D}. \quad (43)$$

Seeing that the conserved quantity is here rc instead of c , the methodology from Section 3.2.2 can be applied (substitute $\Gamma \rightarrow \Gamma_{1D}$ and $\Omega \rightarrow \Omega_{2D}$, $c \rightarrow cr$). This leads to the diffuse interface formulation

$$\partial_t(r|\nabla\phi|c) + \nabla \cdot (r|\nabla\phi|c\mathbf{v}) = D\nabla \cdot (r|\nabla\phi|\nabla c \cdot \mathbf{n}_\phi \otimes \mathbf{n}_\phi) \quad \text{in } \Omega_{2D}. \quad (44)$$

Initial conditions are again $c = 1$ and $c_{1D} = 1$. Further we combine Eqs. (36) and (41) to formulate the surface tension of the elastic force as

$$\sigma(c) = (K_A + K_S) \left(\frac{1}{c_{1D}} - 1 \right) + (K_A - K_S) \left(\frac{c_{1D}}{c} - 1 \right) + \sigma_0.$$

Using Eq. (35) a straightforward diffuse interface version of the force then reads

$$\mathbf{f}_{stretch} = -c \nabla \cdot (\sigma(c) \nabla \phi \otimes \nabla \phi) + 2K_S |\nabla \phi| \begin{pmatrix} 0 \\ 1/r \end{pmatrix} \begin{pmatrix} c - c_{1D}^2 \\ c c_{1D} \end{pmatrix}. \quad (45)$$

Including the force (45) in the axisymmetric Navier–Stokes-equations as a source term gives

$$\rho(\phi) \partial_t^* \mathbf{v} - \nabla \cdot (\eta(\phi) (\nabla \mathbf{v} + (\nabla \mathbf{v})^T)) + \nabla p - \frac{\eta(\phi)}{r} \begin{pmatrix} \partial_r v_x + \partial_x v_r \\ 2\partial_r v_r - \frac{2}{r} v_r \end{pmatrix} = \mathbf{f}_{stretch} \quad \text{in } \Omega_{2D}, \quad (46)$$

$$\nabla \cdot \mathbf{v} + \frac{1}{r} v_r = 0 \quad \text{in } \Omega_{2D}. \quad (47)$$

Hence, the full model of an elastic axisymmetric surface system consists of Eqs. (42) and (44)–(47), coupled to the axisymmetric Cahn–Hilliard-equations

$$\partial_t^* \phi = M(\Delta \mu + \frac{1}{r} \partial_r \mu) \quad \text{in } \Omega_{2D}, \quad (48)$$

$$\mu = \frac{1}{\epsilon} W'(\phi) - \epsilon \left(\Delta \phi + \frac{1}{r} \partial_r \phi \right) \quad \text{in } \Omega_{2D}. \quad (49)$$

3.3.3. Extension to viscoelastic surfaces

A formulation of axisymmetric surface elasticity could also be derived from the evolution of the in-plane surface stress tensor, similar to our approach in Section 3.2.3. This procedure also enables the extension of the equations to describe a viscoelastic surface. Acknowledging the axisymmetric structure of the problem the viscoelastic surface stress tensor can be decomposed as

$$\mathbf{S} = \begin{bmatrix} S_1 P & 0 \\ 0 & S_2 \end{bmatrix}, \quad (50)$$

where S_1 is the scalar longitudinal stress, S_2 is the scalar radial stress and P is still the 2×2 projection matrix on the surface Γ_{1D} . From Eq. (8) one can deduce two evolution equations for S_1 and S_2 and as shown in Appendix B reformulate them in terms of c and c_{1D} to obtain evolution equations, similar to the previous Eqs. (39) and (40), but with additional terms to include viscoelastic relaxation.

The resulting equations read

$$\partial_t(r|\nabla\phi|c) + \nabla \cdot (r|\nabla\phi|c\mathbf{v}) = D\nabla \cdot (r|\nabla\phi|\nabla c \cdot \mathbf{n}_\phi \otimes \mathbf{n}_\phi) + \frac{r}{\tau_A} |\nabla\phi|(1-c) \quad (51)$$

$$\partial_t(|\nabla\phi|c_{1D}) + \nabla \cdot (|\nabla\phi|c_{1D}\mathbf{v}) = D\nabla \cdot (|\nabla\phi|\nabla c_{1D} \cdot \mathbf{n}_\phi \otimes \mathbf{n}_\phi) + \left(\frac{1}{2\tau_A} - \frac{1}{2\tau_S} \right) |\nabla\phi|(1-c) + \frac{1}{\tau_S} |\nabla\phi|(1-c_{1D}), \quad (52)$$

in Ω_{2D} . Eqs. (51) and (52) coupled to Eqs. (46)–(49) provide the complete model to simulate an axisymmetric viscoelastic diffuse interface in fluid flow.

3.4. Fully three-dimensional surfaces

In the previous sections we have presented three-dimensional models for either dilational or axisymmetric surfaces. For surfaces which fall into neither of these two categories, it is necessary to track the evolution of the principle directions of the surface deformation. This involves a tensor-valued surface PDE, as given in Eq. (8). Since Eq. (8) is not a surface conservation equation, it is not amenable to an equally elegant transformation into a diffuse interface formulation. Instead, the straightforward approach is to approximate surface operators by use of the diffuse projection matrix $P_\phi := I - \mathbf{n}_\phi \otimes \mathbf{n}_\phi$, to introduce a diffuse surface gradient $P_\phi \nabla$ and a diffuse surface rate of deformation tensor $D_\phi := \frac{1}{2} P_\phi (\nabla \mathbf{v} + (\nabla \mathbf{v})^T) P_\phi$. Multiplying Eq. (8) with the approximate delta function $|\nabla \phi|$ and adding the normal diffusion, now component-wise, we can derive a diffuse interface formulation. In case of a purely elastic surface ($\tau_A = \tau_S = 0$) we obtain

$$|\nabla \phi| \left(\partial_t^* S - (P_\phi \nabla \mathbf{v}) S - S (P_\phi \nabla \mathbf{v})^T - 2K_S D_\phi - (K_A - K_S) \text{tr}(D_\phi) P_\phi \right) = D \nabla \cdot (|\nabla \phi| \mathbf{n}_\phi \otimes (\mathbf{n}_\phi \cdot \nabla S))$$

Note, that this formulation is less elegant than the formulations derived in the previous sections. In particular the various matrix products involve hundreds of terms, making an efficient numerical treatment challenging. Also, the conservation of the elastic reference state cannot be assured as no conservation properties are guaranteed.

To couple the stress S to the hydrodynamic system, one can approximate its surface divergence by $P_\phi : \nabla S = [\sum_{i,j} P_{\phi,i,j} \partial_i S_{j,k}]_k$, such that the diffuse version of the stretching force becomes

$$\mathbf{f}_{stretch} = |\nabla \phi| P_\phi : \nabla S.$$

If coupled to the Navier–Stokes–Cahn–Hilliard system we obtain a diffuse interface description of a fully three-dimensional elastic surface with in-plane shear elasticity.

4. Discretization

For discretizing the presented systems of equations a semi-implicit Euler scheme, following [22], is used in time and a finite-element discretization in space. For simplicity we just show the discretization for the case of dilational elastic surfaces (see Eqs. (24)–(29) with $K_B = 0$). The discretization for elastic axisymmetric surfaces (see Section 3.3) is done analogously and differs only in additional axisymmetric terms and an extra (axisymmetric) concentration equation.

4.1. Time discretization

For the time discretization we chose an equidistant time partitioning with time step size Δt . For some time-dependent scalar variable f the time derivative can then be approximated by the discrete time derivative $d_t f^n := \frac{f^n - f^{n-1}}{\Delta t} \approx \partial_t f$ at the n th time step.

Employing the ideas from [22, Sec. 4.1] for the Navier–Stokes–Cahn–Hilliard Eqs. (24)–(27) a stable semi-implicit Euler scheme reads

$$d_t \phi^n = \nabla \cdot (M \nabla \mu^n) - \nabla \phi^{n-1} \cdot \mathbf{v}^n, \quad (53)$$

$$\mu^n = -\epsilon \Delta \phi^n + \epsilon^{-1} (W'(\phi^{n-1}) + W''(\phi^{n-1})(\phi^n - \phi^{n-1})), \quad (54)$$

$$\nabla \cdot \mathbf{v}^n = 0, \quad (55)$$

$$\rho(\phi^{n-1}) (d_t \mathbf{v}^n + \mathbf{v}^{n-1} \cdot \nabla \mathbf{v}^n) = \nabla \cdot (2\eta(\phi^{n-1}) D_\Omega^n) - \nabla p^n - \epsilon \nabla \cdot (\sigma(c^{n-1}) \nabla \phi^n \otimes \nabla \phi^{n-1}), \quad (56)$$

where $D_\Omega^n = \frac{1}{2} (\nabla \mathbf{v}^n + (\nabla \mathbf{v}^n)^T)$ denotes the rate of deformation tensor. Here, the solutions ϕ^{n-1} , $\mathbf{v}^{n-1}$ and c^{n-1} were precomputed in the previous time step and the surface tension σ is defined as in Eq. (28). As shown in [22] solving this fully-coupled system monolithically has a strong stabilizing effect, allowing larger time steps. The surface concentration equation is solved in each time step as

$$d_t (|\nabla \phi^{n-1}| c^n) + \nabla \cdot (|\nabla \phi^{n-1}| c^n \mathbf{v}^{n-1}) = D \nabla \cdot (|\nabla \phi^{n-1}| \nabla c^n \mathbf{n}_\phi^{n-1} \otimes \mathbf{n}_\phi^{n-1}) + D_I \nabla \cdot (|\nabla \phi^{n-1}| \nabla (c^n - c^{n-1}) P_\phi^{n-1}), \quad (57)$$

which only depends on the previously computed solutions ϕ^{n-1} , ϕ^{n-2} , c^{n-1} and $\mathbf{v}^{n-1}$. To increase numerical stability we added an artificial diffusion term with diffusion constant D_I , which acts in tangential direction to the surface. Subtraction of the diffusion from the previous time step ensures that this approach is consistent as $\Delta t \rightarrow 0$.

Note, throughout this work Eqs. (53)–(56) and Eq. (57) were solved monolithically. To reduce the size of the linear system the Navier–Stokes–Cahn–Hilliard system and the concentration equation could be solved sequentially instead.

Remark (Conservation of Elastic Reference State). Recall that conservation of the surface mass of c implies conservation of the elastic reference state (see the remark in Section 3.2.2). The proposed time discretization (57) of the concentration equation ensures this mass conservation on the discrete level as can be seen from the weak form [39]. The weak form of (57) with a test function $\beta \in C^0(\bar{\Omega}) \cap H^1(\Omega)$ is given as

$$\int_{\Omega} |\nabla \phi^{n-1}| c^n \beta - \Delta t \int_{\Omega} (|\nabla \phi^{n-1}| c^n \mathbf{v}^{n-1} - D |\nabla \phi^{n-1}| \nabla c^n \mathbf{n}_\phi^{n-1} \otimes \mathbf{n}_\phi^{n-1}) \cdot \nabla \beta = \int_{\Omega} |\nabla \phi^{n-2}| c^{n-1} \beta.$$

Note that we choose $D_I = 0$ for readability here. Choosing $\beta \equiv 1$ yields

$$\int_{\Omega} |\nabla \phi^{n-1}| c^n = \int_{\Omega} |\nabla \phi^{n-2}| c^{n-1},$$

which is the phase-field equivalent of exact mass conservation

$$\int_{\Gamma^{n-1}} c^n = \int_{\Gamma^{n-2}} c^{n-1}.$$

However, small errors in surface mass might occur and accumulate over time due to adaptive grid refinement and coarsening. To ensure mass conservation for long simulations, an artificial mass correction could be incorporated by changing c^n to $c^n + k_{mass} \frac{\int_{\Omega} (c^0 - c^n) |\phi^{n-1}|}{\int_{\Omega} |\phi^{n-1}|}$ with a small constant k_{mass} after each time step.

4.2. Finite element discretization

In space we discretize the presented models using a finite element approach. Let T_h be a triangulation of the domain Ω and $P_m(T)$, $m \in \mathbb{N}$, denote the space of polynomials of degree less than or equal to m on a triangle $T \in T_h$.

Depending on the phase-field function ϕ we use an adaptive refinement strategy for the mesh around the interface. Coarsening and refinement of the mesh takes place such that the region $\phi \in [0.01, 0.99]$ is resolved by at least five degrees of freedom at any time across the interface. Away from the interface the grid size is fixed to a coarser value Δx .

We use Lagrange- P_1 finite elements for the pressure p and P_2 elements for the other variables. The corresponding finite element spaces are

$$\begin{aligned} Q_h &= \{q \in L_0^2(\Omega) \mid q|_T \in P_1(T)\}, \\ U_h &= \{u \in C^0(\bar{\Omega}) \cap H_0^1(\Omega) \mid u|_T \in P_2(T)\}, \\ Y_h &= \{\psi \in C^0(\bar{\Omega}) \mid \psi|_T \in P_2(T)\}, \\ B_h^a &= \{\beta \in C^0(\bar{\Omega}) \cap H^1(\Omega) \mid \beta|_T \in P_2(T), \beta = a \text{ on } \partial\Omega\}. \end{aligned}$$

Using $\langle \cdot, \cdot \rangle$ to denote the L_2 -inner product on Ω the weak formulations for Eqs. (53)–(57) read as follows.

Navier–Stokes–Cahn–Hilliard system: Find $(\phi^n, \mu^n, p^n, \mathbf{v}^n) \in Y_h \times Y_h \times Q_h \times (U_h)^2$ such that for all $(\psi, \xi, q, \mathbf{u}) \in Y_h \times Y_h \times Q_h \times (U_h)^2$

$$\begin{aligned} \langle d_t \phi^n, \psi \rangle &= \langle -M \nabla \mu^n, \nabla \psi \rangle - \langle \nabla \phi^{n-1} \cdot \mathbf{v}^n, \psi \rangle, \\ \langle \mu^n, \xi \rangle - \langle \epsilon \nabla \phi^n, \nabla \xi \rangle &= \epsilon^{-1} \langle W'(\phi^{n-1}) + W''(\phi^{n-1})(\phi^n - \phi^{n-1}), \xi \rangle, \\ \langle \nabla \cdot \mathbf{v}^n, q \rangle &= 0, \\ \rho(\phi^{n-1}) \langle d_t \mathbf{v}^n, \mathbf{u} \rangle + \langle \mathbf{v}^{n-1} \cdot \nabla \mathbf{v}^n, \mathbf{u} \rangle &= - \langle 2\eta(\phi^{n-1}) D_{\Omega}^n, \nabla \mathbf{u} \rangle + \langle p^n, \nabla \cdot \mathbf{u} \rangle + \left\langle \sigma(c^{n-1}) \mu^n \nabla \phi^{n-1} + |\nabla \phi^{n-1}| P_{\phi}^{n-1} \nabla \sigma(c^{n-1}), \mathbf{u} \right\rangle. \end{aligned}$$

Concentration equation: Find $c^n \in B_h^1$ such that for all $\beta \in B_h^0$

$$\langle d_t(|\nabla \phi^{n-1}| c^n), \beta \rangle - \langle |\nabla \phi^{n-1}| c^n \mathbf{v}^{n-1}, \nabla \beta \rangle = -D \left\langle |\nabla \phi^{n-1}| \nabla c^n \mathbf{n}_{\phi}^{n-1} \otimes \mathbf{n}_{\phi}^{n-1}, \nabla \beta \right\rangle - D_I \left\langle |\nabla \phi^{n-1}| \nabla (c^n - c^{n-1}) P_{\phi}^{n-1}, \nabla \beta \right\rangle.$$

The problems are discretized in the finite element toolbox AMDiS [40–42]. The resulting linear system of equations is solved using the direct solver UMFPAK [43]. Note, to ensure regularity of the induced linear system a small constant was added to the numerical approximation $|\nabla \phi|$ of the surface delta function.

5. Numerical results

In the following we present a series of numerical tests to challenge the derived formulations of the in-plane stretching force. While phase-field simulations of surface tension and bending forces are state-of-the-art, e.g. [18,27], we mostly focus on purely stretchable interfaces in the following to assess the validity and accuracy of our new contribution. That is we set in most parts $K_B = \sigma_0 = 0$. This case is also of practical relevance, as can be seen from Eq. (7): the ratio K_B/K_A scales quadratically with the thickness of the shell material, making contributions of bending stiffness negligible if the shell is very thin (compared to the size of the considered object). Also a constant material surface tension σ_0 is neglected in most of our numerical results, to eliminate its stabilizing effect and to challenge the model in a purely (visco)elastic case.

As one main application of our method we identify the numerical simulation of biological cell surfaces. These surfaces are elastic on short time scales and visco-elastic on larger time scales. Consequently, we test our numerical simulations in the regime of typical cell sizes, i.e., on the micrometer scale. For the fluid dynamics this implies the Stokes regime, which is effectively obtained by setting $\rho = 0$ in the momentum equation.

5.1. Toy example: A rubber band in shear flow

As a first experiment to illustrate the conservative properties of our phase-field model we simulate the stretching and relaxation of a rubber band using the 2D model with pure dilational elasticity from Section 3.2.

The rubber band is represented by the interface of the phase-field ϕ and has initially an elliptic shape (axes with length $6\mu\text{m}$ and $4\mu\text{m}$ respectively) centered in the domain $\Omega_{2D} = [-20, 20]\mu\text{m} \times [-10, 10]\mu\text{m}$, Fig. 3 (shape A). We apply a shear flow with magnitude of 1 m s^{-1} through fitting Dirichlet boundary conditions. In consequence, the band stretches over time (shape B). Shear forces are eventually balanced by elastic forces such that the surface tends towards a stationary shape (shape C). After $50\mu\text{s}$ simulation time, we stop the surrounding shear flow. Since no outer force is applied on the rubber band anymore, it starts relaxing and the inherent elastic force drives the band back towards its original length (shapes D and E). The elastic reference state is reached after $150\mu\text{s}$. The final shape is rotated in comparison to the original state (shape F), which is physically sound and illustrates the frame invariance of the elasticity model.

The shapes are colored based on the surface concentration c as a measure of stretching. Initially $c = 1$, such that regions where later $c \neq 1$ holds represent regions of stretching ($c < 1$) and compression ($c > 1$). The bounds of c are also tagged for each snapshot in separate color bars.

Fig. 3 (bottom) shows the evolution of the surface perimeter over the whole process (black curve). For the stationary shape (C) the perimeter reaches a maximum with a stretch of approximately 12%. This is also confirmed by the values of the concentration c for snapshot (C) that implies a stretching of around 13% ($c_{\min} = 0.867$) along most parts of the interface. After switching off the shear flow the surface length nicely converges towards the initial length as expected for an elastic surface. Note, although the phase-field model ensures mass conservation even on the fully discrete level as shown in Section 4.1, due to adaptive grid refinement and coarsening small errors in the surface mass occur and accumulate over time. That is why the surface does not perfectly converge back to its initial length, but a slightly stretched state ($< 0.5\%$) is reached. Note, that these results are obtained on purpose without any mass correction terms, which would lead to perfect recovery of the initial length (just set $k_{\text{mass}} > 0$).

Moreover, we demonstrate the applicability of the viscoelastic Maxwell model described in Section 3.2.3. In Fig. 3 (bottom, colored curves) the evolution of the perimeter of viscoelastic surfaces with relaxation times $\tau \in \{5\mu\text{s}, 10\mu\text{s}, 100\mu\text{s}\}$ are shown. As expected the viscoelastic surface stretches further in comparison to the purely elastic case and does not converge to a stationary state. After switching off the shear flow, the surface does not relax to the initial state but to its new reference state. Note, that the behavior of a viscoelastic surface will tend towards the purely elastic case as $\tau \rightarrow \infty$.

5.2. Numerical validation by comparison with ALE benchmark results

In the following we will validate the phase-field model by comparing results to reference data given in [7]. We simulate the flow of an initially spherical elastic shell through a fluid-filled channel. We consider both, the channel and the shell, to be axisymmetric, see Fig. 4 for an illustration. As described in Section 3.3 axisymmetry effectively reduces the problem to a two-dimensional flow with axisymmetric operators. Consistent with [7] we choose the computational domain as a cylindrical channel of radius $10.94\mu\text{m}$ represented by $\Omega_{2D} = [0, 40]\mu\text{m} \times [0, 10.94]\mu\text{m}$, where the lower boundary represents the symmetry axis. Note, that the channel radius $R = 10.94\mu\text{m}$ mimics the flow through a squared channel with edge length of $20\mu\text{m}$, according to the principle of equivalent channel radii [6]. The simulation setup stems from a modern technique to measure the elastic properties of cell surfaces, called Real-Time-Deformability Cytometry (RT-DC) [5]. In RT-DC experiments, camera snapshots are taken at the end portion of the channel where the cells assume a stationary shape.

Overall, we choose the set of parameters equal to [7] which allows a substantial comparison to the well-validated results of the ALE-approach presented in [7,13]. For comparable results to the ones in [7] we also include an additional constant surface tension term $\sigma_0 = 0.1K_A$. Further, since $\nu = \frac{1}{2}$, it holds $K_S = \frac{1}{3}K_A$, such that the membrane elasticity is ultimately determined by a single free parameter K_A .

The laminar fluid flow with magnitude 0.2 m s^{-1} is induced through appropriate Dirichlet boundary conditions for the velocity field $\mathbf{v}$. Moreover, we impose no slip boundary conditions ($\mathbf{v} = 0$) on the channel wall and no flux boundary conditions ($v_r = 0$) across the symmetry axis.

5.2.1. Benchmark quantity

While exposed to the channel flow, the shell is deformed due to pressure and shear forces. In a sufficiently long channel it will eventually assume a stationary shape whereupon its movement becomes purely translational. We aim in particular to reproduce this state of stationary deformation. As a benchmark quantity to validate our novel phase-field model and numerically evaluate the results of our simulations we use the deformation d as a measure of the deviation of the cell shape from a circle [7,23]. Namely, d is defined as

$$d = 1 - \text{circularity} = 1 - \frac{2\sqrt{A\pi}}{P},$$

where A and P denote the area and perimeter of the 2D view of the deformed cell object, measured by a piecewise linear approximation of the level-set $\phi = 0.5$ of the phase-field function.

In [7] some reference values for the stationary cell shapes for a range of elastic moduli and cell sizes were computed using an ALE-method. There, a co-moving grid was used to keep the cell in the center of the computational domain fixed throughout

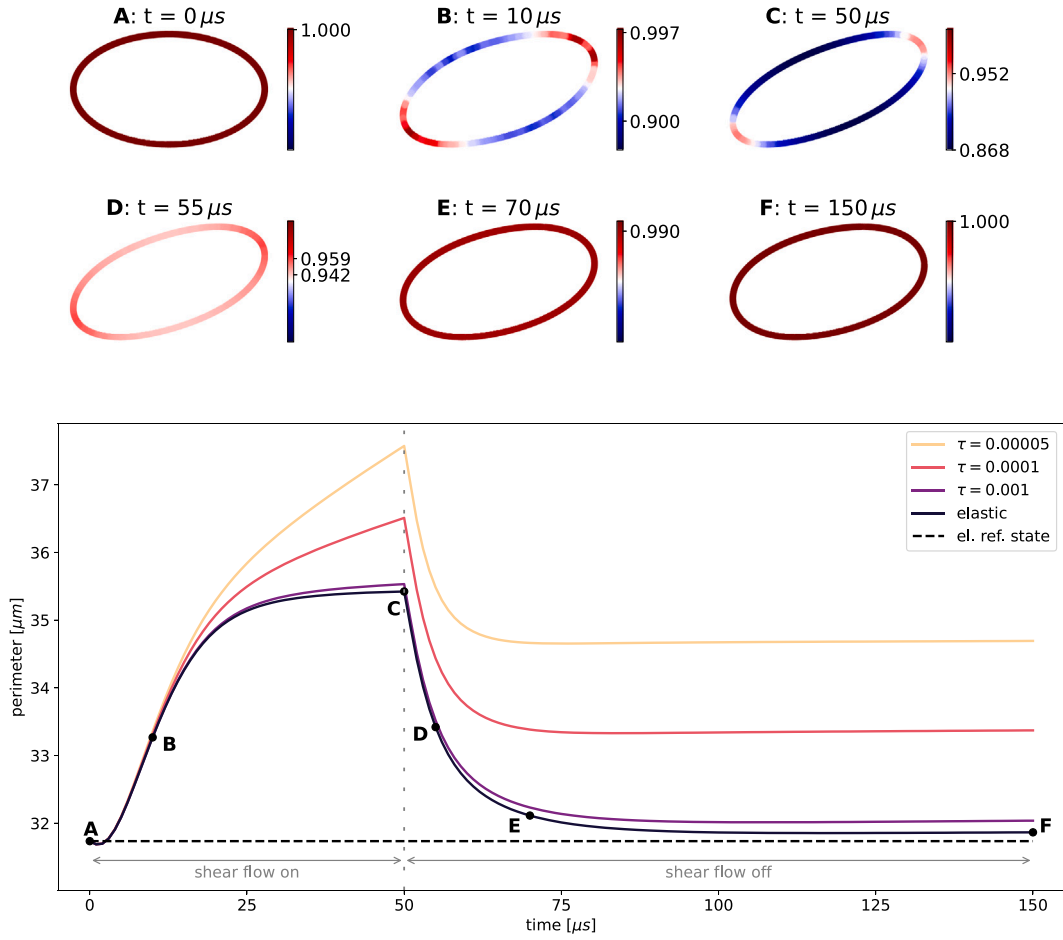


Fig. 3. Simulation of stretching and relaxation of an elastic rubber band in shear flow. **Top:** Six snapshots colored by c as the measure of stretching. All six color bars are scaled equally, but maximal and minimal value of c within each plot are tagged. **Bottom:** Perimeter of the rubber band over time, with additional evolutions of 3 viscoelastic cases ($\tau > 0$). Further Parameters: $K_A = 0.25 \text{ N m}^{-1}$, $\sigma_0 = 0 \text{ N m}^{-1}$, $\epsilon = 3\sqrt{2}e - 7 \text{ m}$, $M = 3\sqrt{2}e - 15 \text{ m}^3 \text{ s}^{-1}$, $D = 2.5e - 6 \text{ m}^2 \text{ s}^{-1}$, $D_I = 1.0e - 7 \text{ m}^2 \text{ s}^{-1}$, $k_{mass} = 0$, $\Delta t = 0.1 \mu\text{s}$, $\Delta x = 0.5 \mu\text{m}$.

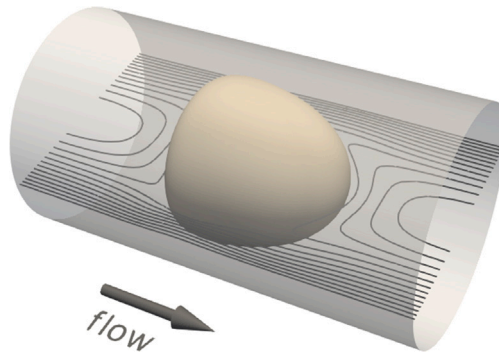


Fig. 4. Stationary shape of an elastic surface flowing through a cylindrical channel. The initially spherical cell surface is deformed by shear and pressure forces to a bullet shape.

the simulation. To match this setup, we also fix the position of the cells in the center of the domain by computing the intrinsic cell velocity $\mathbf{v}_{cell} = (\int_{\Omega_{2D}} \phi \mathbf{v} dx) / (\int_{\Omega_{2D}} \phi dx)$ and subtracting it from the fluid velocity $\mathbf{v}$ within the convective terms of the concentration Eqs. (42) and (44) and the Cahn–Hilliard-equation (48). The numerical results presented in [7] turned out to be highly accurate in terms of spatial and temporal discretization errors. Due to very good agreements with the analytical model shown in [6]

as well as experimental measurements in a validation study using purely elastic spherical particles, the simulation data has been widely used for comparisons to corresponding experiments.

5.2.2. Simulation results

In this section we compare the results of the presented phase-field model with the results of the ALE-approach from [7]. Fig. 5 shows the evolution of the deformation for different sets of parameters. In Fig. 5(a) we consider the dependence on the interface width ϵ . As expected the deformation tends for smaller values of ϵ towards the sharp interface ALE reference solution. Note, the sharp interface limit would not be reached for $\epsilon \rightarrow 0$, unless the mobility constant M is decreased proportional to ϵ [23]. As can be seen from Fig. 5(b) for a fixed interface width (here $\epsilon = 1.5\sqrt{2}e-7$ m) the mobility has almost no influence on the deformation. Note, that a much larger mobility would still lead to a decrease in deformation, since the mobility controls the intrinsic Cahn–Hilliard dynamics that aims to reduce the surface area [23]. As a third experiment we investigate the influence of the normal diffusion D on the deformation. As can be seen from Fig. 5(c) this third model parameter has almost no influence on the deformation in the stationary state as well. Since neither of the model parameters has a significant influence on the deformation in the stationary state our choice of ϵ , M and D can be considered reasonable.

Finally, we compare the results of our novel phase-field approach with the ALE results from [7] over a range of cell sizes and elastic moduli. We simulated cells of different sizes with initial radii in the range of $r \in [5.05 \mu\text{m}, 8.56 \mu\text{m}]$ and with six different values of K_A , chosen in comparison to [7, Fig. 3]. In Fig. 5(d) the deformations of these cells in the steady state for both models are shown. Over a wide range of parameters we get an excellent agreement for both approaches. Although for bigger deformations the results differ more, which can be expected due to the inherent linearity of the models.

The runtime of our validation simulations ranges from approximately 28 to 60 hours (on a single CPU core). This is about 10 times slower than the reference simulations in [7], since the phase-field model requires a much finer resolution at the interface. The distinct advantage of the phase-field approach over the ALE model lies in its flexibility as we will show in the following sections.

5.3. Beyond ALE: Channel traversal without remeshing

In this section we perform further simulation studies in order to illustrate the potential of the presented phase-field model. A major advantage of the phase-field approach is, that due to the implicit interface capturing by the function ϕ , complex deformations can be described. Note that for elastic surfaces complex deformations often times are achieved at small strains, such that these can be accurately described even by linear elastic models. Grid based methods, such as the ALE-approach, track the interface explicitly through the grid. In these methods strong or complex deformations can usually only be handled by continuous re-meshing. To illustrate the power of the phase-field approach we simulate an elastic surface traversing a cylindrical channel without re-meshing. Such simulations are used in practice to understand the dynamics of cell deformation and provide a means to extract cytoplasmic viscosity from microfluidics experiments [44,45].

Fig. 6 shows five significant states of a cell flowing through a channel with conical inlet/outlet of 45° . The flow was again induced through Dirichlet boundary conditions for the velocity, such that within the cylindrical part of the channel a maximum velocity of $v_x = 0.21 \text{ m s}^{-1}$ holds. The deformation depending on the channel position is also shown in Fig. 6. At first the undeformed cell (A) starts moving due to the surrounding flow. Strong forces at the channel entry squeeze the cell and lead to a significant deformation (B). After entry, cell deformation relaxes towards a stationary state (C). When approaching the channel outlet, the cell gets compressed in flow direction. This leads to a small decrease followed by a large increase in deformation. At the maximum deformation (D) the cell has transitioned from its prolate shape in the channel to an oblate shape. After fully leaving the channel the cell starts relaxing (E) back towards its spherical ground state.

5.4. Topological change

In a second example we show that even topological changes, which cannot be simulated with any grid-based approach, can be performed using the phase-field method. Topological changes might for example arise when two vesicles fuse together or during cell division.

Here, we simulate the fusion of two axisymmetric vesicles with dilational elasticity (from Section 3.2). To induce the fusion process of the two vesicles, we use an extensional flow field of the form $\mathbf{v} = (-v_x, \frac{1}{2}v_r)^T$, which is imposed by a Dirichlet boundary condition to $\mathbf{v}$. In this simulation two vesicles with radius $r = 5 \mu\text{m}$ are placed within a distance of $1.6 \mu\text{m}$. We use the same surface dilation modulus $K_A = 0.28 \text{ N m}^{-1}$ for both vesicles. This value was found as the mean value of DOPC vesicles of different sizes in compression experiments [46]. Also we add an additional surface tension of $\sigma_0 = 0.1 K_A$.

Fig. 7 (top) shows the two vesicles in their relaxed state ($c = 1$ everywhere) as well as the stream lines of the surrounding fluid flow at the beginning of the simulation. Both vesicles get slightly stretched ($c < 1$) and start moving towards the center of the simulation domain. After contact, the vesicles begin to fuse. The resulting single vesicle gets horizontally compressed and vertically stretched by the flow. This goes along with areas of strong compression ($c > 1$) around the contact line. Towards the end of the simulation, due to the acting elastic force and the surrounding flow field, the strongly compressed parts of the surface relax and the fused vesicle approaches a slightly compressed stationary state.

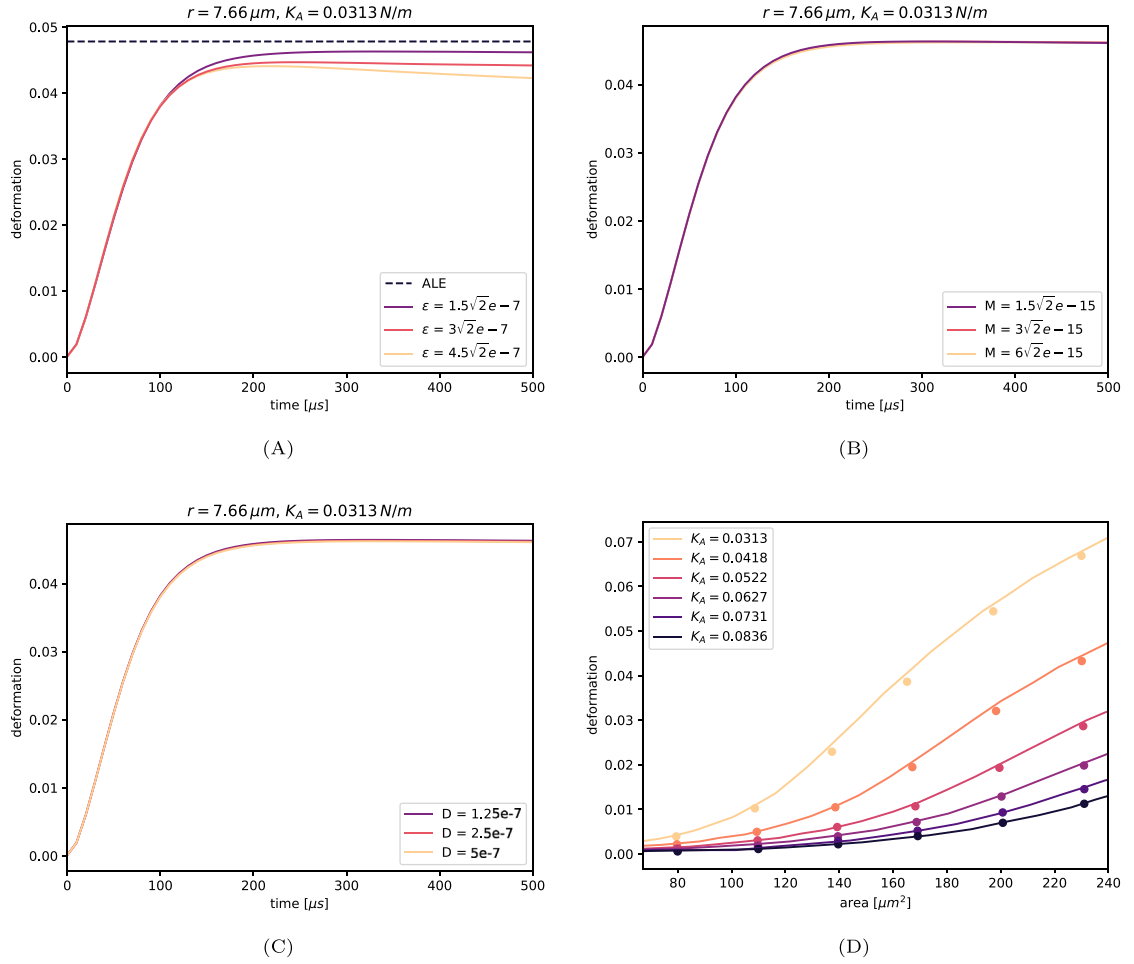


Fig. 5. Deformation of simulated elastic shells in comparison to ALE reference values for varying parameters. In (A)–(C) we use fixed physical parameters with $r = 7.66 \mu\text{m}$ and $K_A = 0.0313 \text{ N m}^{-1}$. The model parameters are $\epsilon = 1.5\sqrt{2}e-7 \text{ m}$, $M = 3\sqrt{2}e-15 \text{ m}^3 \text{ s}^{-1}$ and $D = 2.5e-7 \text{ m}^2 \text{ s}^{-1}$ by default. One of these parameters is varied to investigate the influence of (A) the interface thickness ϵ , (B) the mobility M and (C) the normal diffusion D . (D): Isolines in area-deformation space for linear elastic shells. Lines mark the ALE results, marker points denote the results of the novel phase-field approach. Numerical parameters: $\Delta x = 0.5 \mu\text{m}$, $\Delta t = 0.1 \mu\text{s}$, $D_I = 1e-8 \text{ m}^2 \text{ s}^{-1}$, $k_{\text{mass}} = 0.1$.

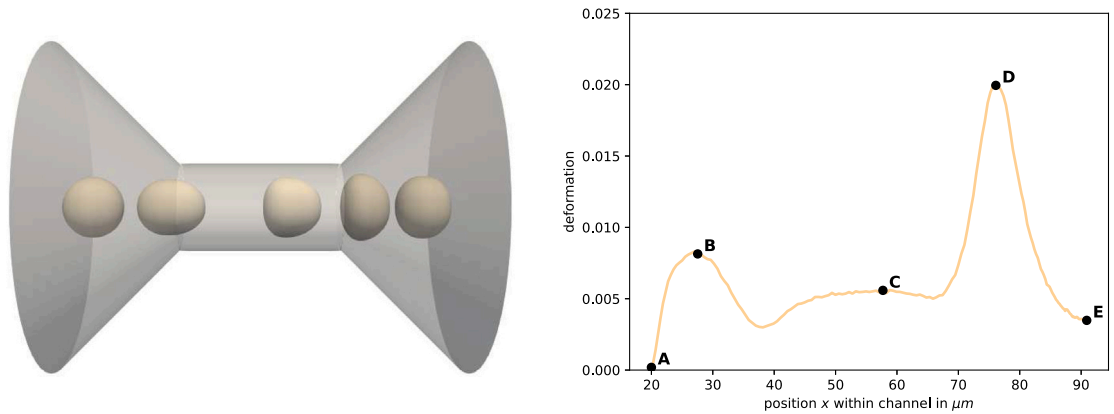


Fig. 6. Simulation of a single cell flowing through a cylindrical channel including inflow/outflow through conical inlet/outlet. Left: Cell shape at five distinct positions within the channel. Right: Cell deformation over channel position. The distinct positions from (left) are labeled A–E. Parameters: $K_A = 0.1 \text{ N m}^{-1}$, $\sigma_0 = 0 \text{ N m}^{-1}$, $r = 7.66 \mu\text{m}$, $\epsilon = 24\sqrt{2}e-7 \text{ m}$, $M = 3\sqrt{2}e-14 \text{ m}^3 \text{ s}^{-1}$, $D = 2.5e-7 \text{ m}^2 \text{ s}^{-1}$, $D_I = 1.0e-8 \text{ m}^2 \text{ s}^{-1}$, $k_{\text{mass}} = 0.1$, $\Delta x = 1.4 \mu\text{m}$ and $\Delta t = 0.4 \mu\text{s}$.

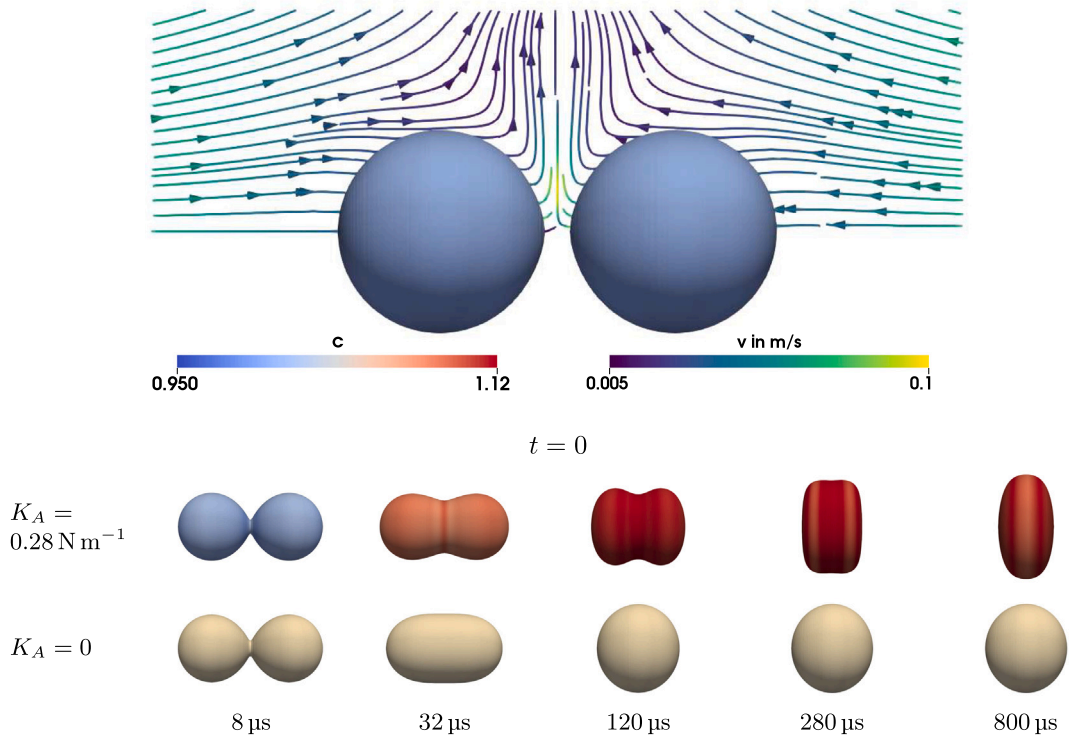


Fig. 7. Simulation of coalescence of two vesicles in extensional flow for elastic surface ($K_A = 0.28 \text{ N m}^{-1}$) and inelastic surface ($K_A = 0$). In the elastic case, snapshots of the fusion process are colored by c as a measure of local compression, constant surface tension without surface elasticity is colored in beige. Parameters: $r = 5.0 \mu\text{m}$, $K_S = 0.0 \text{ N m}^{-1}$, $\epsilon = 24\sqrt{2}e - 7 \text{ m}$, $M = 3\sqrt{2}e - 14 \text{ m}^3 \text{ s}^{-1}$, $D = 2.5e - 6 \text{ m}^2 \text{ s}^{-1}$, $D_I = 1.0e - 7 \text{ m}^2 \text{ s}^{-1}$, $\Delta x = 0.5 \mu\text{m}$, $\Delta t = 0.4 \mu\text{s}$, $\sigma_0 = 0.028 \text{ N m}^{-1}$, $k_{\text{mass}} = 0.1$.

Remark. We note that due to the continuum description of the surface, various molecular steps which happen during the topological change are neglected (e. g. lipid rotation, stalk, hemifusion), assuming that their energetic contributions are highly local and hence negligible for the overall mechanics of the process. Also note, that the numerical simulation of fusion only makes sense for surfaces without in-plane shear elasticity. Otherwise high shear stresses would inhibit any growth of the initial fusion pore (i. e. the ring-shaped area connecting both surfaces). This problem does not arise for dilational surfaces, such as lipid vesicles since the lipids can freely rearrange during the growth of the fusion pore.

For comparison we repeat the same simulation without surface elasticity but only constant surface tension. In this case fundamentally different shape dynamics occur (see Fig. 7, bottom row). Right after contact the vesicles start reducing their surface area to minimize the surface energy until eventually an almost spherical stationary shape is reached.

6. Summary

In this paper, we presented a first phase-field model for elastic and viscoelastic hyper-surfaces in Navier–Stokes fluids, which includes in-plane stretching. The model is based on a coupling of a phase-field model for two-phase flow to a fully Eulerian description of the surface deformation tensor. We put a particular focus on two cases: purely dilational surface elasticity and 3D axisymmetric surface elasticity. For these special cases, we proposed methods which track the evolution of in-plane strains and stresses by simple *scalar* surface PDEs. The governing equations are derived from two different approaches, either from variation of the elastic energy, or from evolution of a viscoelastic surface stress [32]. In the absence of viscous stress dissipation, we obtain equal results for both approaches which validates the modeling. The obtained surface PDEs can be localized to the diffuse interface by the use of an approximate surface delta function, maintaining conservation properties for the elastic reference state (see the remarks in Sections 4.1 and 3.2.2). Bending stiffness can or cannot be included depending on the properties of the underlying physical system.

To validate the applicability and accuracy of the new phase-field model we conducted several numerical experiments. As a benchmark scenario we considered a single elastic cell traversing a fluid channel and compared the results to reference values from an ALE approach [7]. We found very good agreement for various cell sizes and elastic moduli. In particular, we showed that the mobility M and the additional normal diffusion constant D are small enough to influence the results only marginally. Furthermore, we showed that the diffuse interface results tend towards the sharp interface reference solutions for decreasing interface thickness ϵ .

Finally, we highlighted some distinct advantages of the new model as compared to traditional interface tracking methods. We demonstrated the movement of a elastic shell through a fluidic channel without grid re-triangulation. Also, we showed the ability to perform topological changes, while ensuring conservative properties.

While we restricted the modeling to linear and (neo-)Hookean surface materials the model can easily be adapted for other hyperelastic material laws. Also, due to the linearly implicit discretization, the approach does not need iterative solvers and can be easily coupled with phase-field models of different bulk materials [23]. We therefore believe that the proposed phase-field model is well suited to tackle a range of complicated problems such as surface fluidization and solidification, or problems involving the mechanics of biological membranes, in the future.

CRedit authorship contribution statement

Maximilian Kloppe: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Sebastian Aland:** Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

SA acknowledges support from the German Research Foundation (grants AL1705/3-2 and AL1705/6). The authors gratefully acknowledge computing time on the high-performance computer at the NHR Center of TU Dresden. This center is jointly supported by the Federal Ministry of Education and Research and the state governments participating in the NHR (www.nhr-verein.de/unsere-partner). Also the authors acknowledge computing time on the compute cluster of the Faculty of Mathematics and Computer Science of Technische Universität Bergakademie Freiberg, operated by the computing center (URZ) and funded by the Deutsche Forschungsgemeinschaft (DFG) under DFG grant number 397252409.

Appendix A. Asymptotic analysis

In this section we argue that the solution of the phase-field concentration Eq. (23) remains continuous and constant in normal direction across the interface, while maintaining the correct sharp interface limit. Specifically, using matched asymptotic analysis we show that for $\epsilon \rightarrow 0$ the diffuse interface equation

$$\partial_t(|\nabla\phi|c) + \mathbf{v} \cdot \nabla(|\nabla\phi|c) = D\nabla \cdot (|\nabla\phi|\nabla c \cdot \mathbf{n} \otimes \mathbf{n}) \quad (\text{A.1})$$

converges formally to Eq. (21) and

$$\mathbf{n} \cdot \nabla c = 0 \quad \text{on } \Gamma \quad (\text{A.2})$$

holds. To do so we follow the argumentation in [39,47–49].

First, a new inner coordinate system is introduced in a neighborhood of Γ , such that for any x in the neighborhood there is a unique $s(x) \in \Gamma$ with minimal distance to x . Now one can write

$$x = s(x) + r(x)\mathbf{n} = s(x) + \epsilon z(x)\mathbf{n},$$

with $r(x)$ being a signed distance function with $r < 0$ in Ω_1 and $r > 0$ in Ω_0 and $z(x) = \frac{r(x)}{\epsilon}$ being a scaled distance function. The phase-field function expressed in these new coordinates then reads

$$\phi = \frac{1}{2} \left(1 - \tanh \left(\frac{3r}{\epsilon} \right) \right) = \frac{1}{2} (1 - \tanh(3z)). \quad (\text{A.3})$$

Outside of the interface region an outer expansion $c = \sum_{i=0}^{\infty} \epsilon^i c^{(i)}$ in terms of the original coordinate system is introduced. Within the neighborhood of Γ we introduce an inner expansion of the concentration field, $\hat{c} = \sum_{i=0}^{\infty} \epsilon^i \hat{c}^{(i)}$, in terms of the new coordinate system. By matching both expansions in an overlapping region, where both are valid representations of the same function, the following matching conditions hold

$$\lim_{r \rightarrow \pm 0} c^{(0)}(s, r) = \lim_{z \rightarrow \pm \infty} \hat{c}^{(0)}(s, z), \quad (\text{A.4})$$

$$\lim_{r \rightarrow \pm 0} \nabla c^{(0)}(s, r) \cdot \mathbf{n} = \lim_{z \rightarrow \pm \infty} \partial_z \hat{c}^{(1)}(s, z). \quad (\text{A.5})$$

Analogously an outer expansion $\mathbf{v} = \sum_{i=0}^{\infty} \epsilon^i \mathbf{v}^{(i)}$ and an inner expansion $\hat{\mathbf{v}} = \sum_{i=0}^{\infty} \epsilon^i \hat{\mathbf{v}}^{(i)}$ of the velocity field $\mathbf{v}$ are defined and the matching condition

$$\lim_{r \rightarrow \pm\infty} \mathbf{v}^{(0)}(s, r) = \lim_{z \rightarrow \pm\infty} \hat{\mathbf{v}}^{(0)}(s, z) \quad (\text{A.6})$$

holds. Inserting the outer expansions into Eq. (A.1) yields to zeroth order

$$0 = \partial_t (|\nabla\phi|c^{(0)}) + \mathbf{v}^{(0)} \cdot \nabla (|\nabla\phi|c^{(0)}) - D\nabla \cdot (|\nabla\phi|\nabla c^{(0)} \cdot \mathbf{n} \otimes \mathbf{n}) \quad (\text{A.7})$$

which gives the trivial identity $0 = 0$, since $|\nabla\phi| = 0$ holds away from the interface.

To transform Eq. (A.1) into the inner coordinate system (s, z) we use the following identities for a scalar function $b(t, x)$. It holds

$$\partial_t b = \partial_t^\circ \hat{b} - \frac{1}{\epsilon} v_r \partial_z \hat{b} + \mathcal{O}(\epsilon), \quad (\text{A.8})$$

$$\nabla b = \left(\frac{1}{\epsilon} \mathbf{n} \partial_z + \nabla_r \right) \hat{b}, \quad (\text{A.9})$$

where $\hat{b}(t, s(t, x), z(t, x)) = b(t, x)$ is the denotation of b in the new coordinates, v_r is the normal velocity of the interface (0.5 level set of ϕ) and $\partial_t^\circ(\cdot) = \partial_t(\cdot) + v_r \mathbf{n} \cdot \nabla(\cdot)$ denotes the normal time derivative, see e. g. [35,49]. Using these equations as well as $\mathbf{n} \cdot \nabla_r = 0$ and $\partial_z \mathbf{n} = 0$ we get

$$0 = \partial_t^\circ (|\nabla\phi|\hat{c}) - \frac{1}{\epsilon} v_r \partial_z (|\nabla\phi|\hat{c}) + \hat{\mathbf{v}} \cdot \left(\frac{1}{\epsilon} \mathbf{n} \partial_z (|\nabla\phi|\hat{c}) + \nabla_r (|\nabla\phi|\hat{c}) \right) - D \left(\frac{1}{\epsilon} \mathbf{n} \partial_z + \nabla_r \right) \cdot \left(|\nabla\phi| \left(\frac{1}{\epsilon} \mathbf{n} \partial_z \hat{c} + \nabla_r \hat{c} \right) \cdot \mathbf{n} \otimes \mathbf{n} \right) + \mathcal{O}(\epsilon) \quad (\text{A.10})$$

$$= \partial_t^\circ (|\nabla\phi|\hat{c}) - \frac{1}{\epsilon} v_r \partial_z (|\nabla\phi|\hat{c}) + \hat{\mathbf{v}} \cdot \nabla_r (|\nabla\phi|\hat{c}) + \frac{1}{\epsilon} \hat{\mathbf{v}} \cdot \mathbf{n} \partial_z (|\nabla\phi|\hat{c}) - \frac{1}{\epsilon} D \nabla_r \cdot (|\nabla\phi| \mathbf{n} \partial_z \hat{c}) - \frac{1}{\epsilon^2} D \partial_z (|\nabla\phi| \partial_z \hat{c}) + \mathcal{O}(\epsilon). \quad (\text{A.11})$$

Now, inserting the inner expansion into Eq. (A.11) and comparing powers of ϵ , gives the following conditions. At **order** $1/\epsilon^2$ it holds

$$0 = -D \partial_z (|\nabla\phi| \partial_z \hat{c}^{(0)}). \quad (\text{A.12})$$

Note, that ϕ as given by Eq. (A.3) does not require an expansion, since $\hat{\phi} \equiv \hat{\phi}^{(0)}$. Since $|\nabla\phi| \rightarrow 0$ for $z \rightarrow \pm\infty$ this yields $\partial_z \hat{c}^{(0)} = 0$. Using this result, we get at **order** $1/\epsilon$

$$0 = -v_r \partial_z (|\nabla\phi| \hat{c}^{(0)}) + \hat{\mathbf{v}}^{(0)} \cdot \mathbf{n} \partial_z (|\nabla\phi| \hat{c}^{(0)}) - D \partial_z (|\nabla\phi| \partial_z \hat{c}^{(1)}). \quad (\text{A.13})$$

We note that $\hat{\mathbf{v}}^{(0)} \cdot \mathbf{n}$ is constant in z and the normal velocity v_r is continuous across the interface, which can be seen by performing an asymptotic expansion of the velocity field in the balance of mass equation and applying matching condition Eq. (A.6) (see [49, Eqs. (4.12), (4.13) and (4.16)]). With that, the first two terms on the right hand side cancel each other out and integration of Eq. (A.13) over z shows analogously $\partial_z \hat{c}^{(1)} = 0$. Applying matching condition (A.5) gives

$$\mathbf{n} \cdot \nabla c^{(0)} = 0 \quad \text{on } \Gamma$$

and with that Eq. (A.2). Finally, using $\partial_z \hat{c}^{(1)} = 0$ and again $\partial_z (\hat{\mathbf{v}}^{(0)} \cdot \mathbf{n}) = 0$ as well as the continuity of the normal velocity, we obtain at **order** 1

$$0 = \partial_t^\circ (|\nabla\phi| \hat{c}^{(0)}) + \hat{\mathbf{v}}^{(1)} \cdot \mathbf{n} \partial_z (|\nabla\phi| \hat{c}^{(0)}) + \hat{\mathbf{v}}^{(0)} \cdot \nabla_r (|\nabla\phi| \hat{c}^{(0)}) - D \partial_z (|\nabla\phi| \partial_z \hat{c}^{(2)}). \quad (\text{A.14})$$

From the balance of mass equation one also gets

$$\partial_z (\hat{\mathbf{v}}^{(1)} \cdot \mathbf{n}) + \nabla_r \cdot \hat{\mathbf{v}}^{(0)} = 0, \quad (\text{A.15})$$

see [49, Eq. (4.15)]. Integrating Eq. (A.14) from $z = -\infty$ to $z = \infty$ gives for the first term

$$\int_{-\infty}^{\infty} \partial_t^\circ (|\nabla\phi| \hat{c}^{(0)}) dz = \int_{-\infty}^{\infty} (\hat{c}^{(0)} \partial_t^\circ |\nabla\phi| + \partial_t^\circ \hat{c}^{(0)} |\nabla\phi|) dz = \partial_t^\circ \hat{c}^{(0)} \int_{-\infty}^{\infty} |\nabla\phi| dz,$$

where we used that $\hat{c}^{(0)}$ is constant in z and an inner expansion of ϕ would show that at order 1 the phase-field function ϕ does not depend on t (see e. g. [23, Eq. (3.14)]). Partial integration of the second term of Eq. (A.14) and applying condition (A.15) results in

$$\begin{aligned} \int_{-\infty}^{\infty} \hat{\mathbf{v}}^{(1)} \cdot \mathbf{n} \partial_z (|\nabla\phi| \hat{c}^{(0)}) dz &= [\hat{\mathbf{v}}^{(1)} \cdot \mathbf{n} |\nabla\phi| \hat{c}^{(0)}]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \partial_z (\hat{\mathbf{v}}^{(1)} \cdot \mathbf{n}) |\nabla\phi| \hat{c}^{(0)} dz \\ &= 0 + \int_{-\infty}^{\infty} \nabla_r \cdot \hat{\mathbf{v}}^{(0)} |\nabla\phi| \hat{c}^{(0)} dz \\ &= \hat{c}^{(0)} \nabla_r \cdot \hat{\mathbf{v}}^{(0)} \int_{-\infty}^{\infty} |\nabla\phi| dz, \end{aligned}$$

since $\hat{\mathbf{v}}^{(0)}$ is independent of z . Integration of the third term of Eq. (A.14) gives

$$\int_{-\infty}^{\infty} \hat{\mathbf{v}}^{(0)} \cdot \nabla_r (|\nabla\phi| \hat{c}^{(0)}) dz = \int_{-\infty}^{\infty} \hat{\mathbf{v}}^{(0)} \cdot (\hat{c}^{(0)} \nabla_r |\nabla\phi| + |\nabla\phi| \nabla_r \hat{c}^{(0)}) dz = \hat{\mathbf{v}}^{(0)} \cdot \nabla_r \hat{c}^{(0)} \int_{-\infty}^{\infty} |\nabla\phi| dz,$$

since ϕ does not depend on s as can be seen from Eq. (A.3). The fourth term of Eq. (A.14) vanishes with integration as $|\nabla\phi| \rightarrow 0$ for $z \rightarrow \pm\infty$ holds. With that integration of Eq. (A.14) leads to

$$0 = \partial_t^2 \hat{c}^{(0)} \int_{-\infty}^{\infty} |\nabla\phi| dz + \hat{c}^{(0)} \nabla_F \cdot \hat{\mathbf{v}}^{(0)} \int_{-\infty}^{\infty} |\nabla\phi| dz + \hat{\mathbf{v}}^{(0)} \cdot \nabla_F \hat{c}^{(0)} \int_{-\infty}^{\infty} |\nabla\phi| dz.$$

Dividing this result by $\int_{-\infty}^{\infty} |\nabla\phi| dz$ and using $\partial_t^2(\cdot) = \partial_t^2(\cdot) + \mathbf{v} \cdot \nabla_F(\cdot)$ leads to

$$0 = \partial_t^2 \hat{c}^{(0)} + \hat{c}^{(0)} \nabla_F \cdot \hat{\mathbf{v}}^{(0)}. \quad (\text{A.16})$$

Finally, using the matching conditions (A.4) and (A.6) we retrieve Eq. (21), showing the formal convergence of the phase-field concentration Eq. (A.1).

Appendix B. Derivation of 3D axisymmetric viscoelastic model

In this section we show how to derive the evolution equations (51) and (52) for axisymmetric viscoelastic surfaces. From Eq. (8) one can deduce two evolution equations for the decomposed surface stress tensor (50). It holds

$$\partial_t^* S_1 = 2K_S \nabla_F \cdot \mathbf{v} + (K_A - K_S) \left(\nabla_F \cdot \mathbf{v} + \frac{1}{r} v_r \right) - \frac{1}{\tau_S} S_1 + \frac{\tau_A - \tau_S}{2\tau_A \tau_S} (S_1 + S_2) + \mathcal{O}(S_1), \quad (\text{B.1})$$

$$\partial_t^* S_2 = 2K_S \frac{1}{r} v_r + (K_A - K_S) \left(\nabla_F \cdot \mathbf{v} + \frac{1}{r} v_r \right) - \frac{1}{\tau_S} S_2 + \frac{\tau_A - \tau_S}{2\tau_A \tau_S} (S_1 + S_2) + \mathcal{O}(S_2). \quad (\text{B.2})$$

Note, that the terms $\mathcal{O}(S_1), \mathcal{O}(S_2)$ are negligible, since we will base the following derivation on the assumption of small stresses, i. e. $S_1, S_2 \ll K_S$, similar to linear elasticity theory.

Eqs. (B.1) and (B.2) are two scalar equations to track the elastic stress on a sharp interface. In principle one could use a phase-field approach of these equations to compute the viscoelastic stress on an axisymmetric diffuse interface. However, this approach is less elegant than the above presented formulation in the variables c and c_{1D} . In particular the diffuse interface formulation would contain many explicit computations of surface divergence operators. Also, S_1 and S_2 are not conserved variables, hence, it could not be ensured that the elastic reference state is exactly maintained on the fully discrete level if relaxation goes to zero (i. e. $\tau_A, \tau_S \rightarrow \infty$). To achieve this, we propose to reformulate the viscoelastic equations in terms of c and c_{1D} . The goal is to obtain evolution equations, similar to the previous Eqs. (39) and (40), but with additional terms to include viscoelastic relaxation. Hence, denoting the relaxation terms by F and G we make the ansatz

$$\partial_t^* c = -c(\nabla_F \cdot \mathbf{v} + \frac{1}{r} v_r) + F, \quad (\text{B.3})$$

$$\partial_t^* c_{1D} = -c_{1D} \nabla_F \cdot \mathbf{v} + G. \quad (\text{B.4})$$

To determine F and G we will connect these evolutions of c, c_{1D} with the given evolutions of S_1, S_2 , respectively. The stress resulting from in-plane deformations of an axisymmetric linear elastic shell is given by [50]

$$S = K_A (\lambda_1 + \lambda_2 - 2) \begin{bmatrix} P & 0 \\ 0 & 1 \end{bmatrix} + K_S \begin{bmatrix} (\lambda_1 - \lambda_2) P & 0 \\ 0 & (\lambda_2 - \lambda_1) \end{bmatrix}. \quad (\text{B.5})$$

Splitting this stress tensor according to Eq. (50) leads to

$$S_1 = (K_A + K_S) \lambda_1 + (K_A - K_S) \lambda_2 - 2K_A, \quad (\text{B.6})$$

$$S_2 = (K_A - K_S) \lambda_1 + (K_A + K_S) \lambda_2 - 2K_A. \quad (\text{B.7})$$

Identifying $\lambda_1 = 1/c_{1D}$, $\lambda_2 = c_{1D}/c$ and computing the time derivative gives

$$\partial_t^* S_1 = \left((K_A - K_S) \frac{1}{c} - (K_A + K_S) \frac{1}{c_{1D}^2} \right) \partial_t^* c_{1D} - (K_A - K_S) \frac{c_{1D}}{c^2} \partial_t^* c, \quad (\text{B.8})$$

$$\partial_t^* S_2 = \left((K_A + K_S) \frac{1}{c} - (K_A - K_S) \frac{1}{c_{1D}^2} \right) \partial_t^* c_{1D} - (K_A + K_S) \frac{c_{1D}}{c^2} \partial_t^* c. \quad (\text{B.9})$$

Combining this with Eqs. (B.3) and (B.4) and using the small strain approximations $c \approx 1$ and $c_{1D} \approx 1$ leads to first order

$$\partial_t^* S_1 = (K_A + K_S) \nabla_F \cdot \mathbf{v} + (K_A - K_S) \frac{1}{r} v_r - 2K_S G - (K_A - K_S) F, \quad (\text{B.10})$$

$$\partial_t^* S_2 = (K_A - K_S) \nabla_F \cdot \mathbf{v} + (K_A + K_S) \frac{1}{r} v_r + 2K_S G - (K_A + K_S) F. \quad (\text{B.11})$$

Without viscoelastic relaxation ($F = G = 0$) we recover the above evolution equations given in Eqs. (B.1) and (B.2). This confirms the validity of the evolution equations to represent elastic surfaces. To include viscoelastic relaxation, we can match the relaxation terms in both evolutions. Solving this for the two unknowns F and G we obtain

$$F = \frac{S_1 + S_2}{2K_A \tau_A}, \quad (\text{B.12})$$

$$G = \frac{S_1 - S_2}{4K_S\tau_S} + \frac{S_1 + S_2}{4K_A\tau_A}. \quad (\text{B.13})$$

Reformulating both equations in terms of c , c_{1D} (using Eqs. (B.6) and (B.7) with $\lambda_1 = \frac{1}{c_{1D}}$, $\lambda_2 = \frac{c_{1D}}{c}$) gives after linearization

$$F = \frac{1}{\tau_A}(1 - c), \quad (\text{B.14})$$

$$G = \left(\frac{1}{2\tau_A} - \frac{1}{2\tau_S} \right) (1 - c) + \frac{1}{\tau_S}(1 - c_{1D}). \quad (\text{B.15})$$

The terms can be easily included in the diffuse interface formulation, leading to Eqs. (51) and (52).

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